ML Architecture Benchmark — Inverse Scattering Results

ECE 228 Final Project | 5 Models | Angle-Conditioned | CPU Training

1. Dataset & Training Configuration

Parameter	Value	Parameter	Value
Total samples	2,000	Input shape X	(2000, 1, 128, 128)
Output shape Y	(2000, 3, 128, 128)	Metadata shape	(2000, 9)
Train / Val / Test split	1600 / 200 / 200	Epochs	30
Input	Single XMCD amplitude pattern I(q)	Output	3-component spin field (Mx, My, Mz)
Angle conditioning	Yes — 9-dim metadata vector per sample	Device	CPU
Sensitivities shape	(2000, 3)	Loss function	MSE

2. Model Architecture Summary

Model	Parameters	Train Time (s)	Train Time (hrs)	Notes
CNN	170,891	7,049	1.96 hr	Plain conv encoder-decoder. Smallest model. Slowest to converge relative to size.
U-Net	1,999,723	5,451	1.51 hr	Skip connections between encoder/decoder. Best accuracy. ~2M params.
Transformer	985,059	1,662	0.46 hr	Self-attention across spatial patches. Fastest to train. Good generalizer.
FNO	7,092,747	3,682	1.02 hr	Fourier Neural Operator. Largest model by far (7M params). Worst overfit.
Neural ODE	175,691	82,575	22.94 hr	ODE solver in the forward pass. Tiny model but catastrophically slow on CPU (23 hrs).

** All models trained on CPU. Neural ODE time dominated by ODE solver calls per forward pass — GPU would reduce this dramatically.*

3. Test Set Results — All Models (ranked by MSE Total)

Rank	Model	MSE Total	MSE Mx	MSE My	MSE Mz	Mean Error Norm	Best Val Loss
1	U-Net	0.2019	0.1253	0.1210	0.3595	0.5920	0.2070
2	FNO	0.2580	0.1253	0.1211	0.5277	0.7842	0.2632
3	Transformer	0.2624	0.1253	0.1211	0.5407	0.7989	0.2706
4	Neural ODE	0.2818	0.1252	0.1210	0.5991	0.8627	0.2909
5	CNN	0.3206	0.1255	0.1210	0.7153	0.9668	0.3244

CRITICAL OBSERVATION: MSE Mx and MSE My are IDENTICAL across all 5 models (Mx ~0.1252-0.1255, My ~0.1210-0.1211). No architecture learned the in-plane spin components. All models predict a near-constant (zero) map for Mx and My. Only Mz varies by architecture, confirming the scattering geometry provides insufficient in-plane information from a single azimuthally-averaged amplitude pattern.

4. Training Dynamics — Final & Best Epoch Values

Model	Ep 1 Train	Ep 30 Train	Ep 1 Val	Ep 30 Val	Best Val	Train/Val Gap	Behavior
CNN	0.3354	0.3200	0.3328	0.3244	0.3244	–0.0044	Slow monotonic decline. Barely learning.
U-Net	0.3371	0.1205	0.3314	0.2149	0.2070	+0.0944	Strong learning. Val lags train — mild overfit. Still improving at ep 30.
Transformer	0.3359	0.2609	0.3332	0.2706	0.2706	–0.0097	Steady convergence. Best train/val agreement. Val still dropping at ep 30.
FNO	0.3275	0.0875	0.3176	0.3054	0.2632	+0.2179	Fast train drop, val plateaus at ep 5. Severe overfit — 0.088 train vs 0.305 val.

Neural ODE	0.3361	0.2734	0.3324	0.2909	0.2909	–0.0175	Slow, noisy val curve. Spike at ep 5 (val=0.353). Reasonable convergence.

5. Per-Sample Reconstruction Quality — Best / Median / Worst Cases

Model	Best MSE (sample idx)	Median MSE (sample idx)	Worst MSE (sample idx)	Best-Worst Spread
CNN	0.2226 (#129)	0.3296 (#96)	0.3528 (#165)	0.1302
U-Net	0.0596 (#170)	0.1846 (#63)	0.4655 (#31)	0.4059
Transformer	0.0802 (#156)	0.2958 (#37)	0.4353 (#171)	0.3551
FNO	0.0709 (#129)	0.2723 (#5)	0.3743 (#16)	0.3034
Neural ODE	0.0913 (#129)	0.3202 (#107)	0.4145 (#125)	0.3232

Notable: Sample #129 is the best case for CNN, FNO, and Neural ODE simultaneously — suggesting certain scattering geometries are inherently easier to invert regardless of architecture. U-Net has the lowest best-case MSE (0.0596) but also the highest worst-case (0.4655) — widest spread of any model.

6. Efficiency Analysis — Accuracy vs. Cost

Model	MSE Total	Params	Train Time	MSE per 1M Params	MSE per 10k sec	Verdict
CNN	0.3206	171K	7,049s / 1.96hr	1.876	0.455	Worst accuracy, mid cost. No advantage.
U-Net	0.2019	2.0M	5,451s / 1.51hr	0.101	0.371	Best accuracy. Reasonable cost. Winner.
Transformer	0.2624	985K	1,662s / 0.46hr	0.266	1.579	Best efficiency. Fastest. Good generalization.
FNO	0.2580	7.1M	3,682s / 1.02hr	0.036	0.701	Huge model, severe overfit. Not justified.
Neural ODE	0.2818	176K	82,575s / 22.9hr	1.603	0.034	Impractical on CPU. Small model, huge time cost.

7. Key Findings

#	Finding
1	U-Net is the best overall model (MSE=0.2019). Skip connections between encoder/decoder levels allow it to preserve spatial structure that other architectures lose.
2	Transformer is the most efficient model (0.46hr training, MSE=0.2624). Val loss still declining at ep 30 — more epochs would likely close the gap with U-Net.
3	FNO severely overfits: train MSE=0.088 vs val=0.305 at ep 30. Its 7M parameters are not justified by the dataset size (1600 training samples).
4	Neural ODE is computationally impractical on CPU (23 hrs). Per-accuracy performance is not competitive. GPU training would be required for fair comparison.
5	CNN completely fails to learn structure: train loss declines only 0.015 over 30 epochs. The plain encoder-decoder without skip connections cannot recover spatial domain patterns.
6	NO model learned Mx or My (identical MSE ~0.125 across all architectures). Only Mz shows architecture-dependent variation. This is a fundamental sensing geometry limitation, not a model capacity issue.
7	Sample #129 is universally easy: best case for CNN, FNO, and Neural ODE. Sample-level MSE variance is large (spread up to 0.41 for U-Net), suggesting geometry-dependent difficulty.
8	All models were trained on CPU with 30 epochs. Results are preliminary — U-Net and Transformer in particular would benefit from GPU training, more epochs, and data augmentation.